

Debjani Nath Sarma

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PROFESSIONAL SUMMARY

Data Analyst with experience delivering enterprise reporting, ETL workflows, and data integration initiatives across banking, retail, and healthcare domain. Skilled in SQL, Python, Power BI, Tableau, and data warehousing with expertise in dashboard development, data validation, reporting automation, and source-to-target mapping.

SKILLS

Business Intelligence and Analytics: SQL (SSMS, MySQL, Oracle DB), Python (Pandas, NumPy, Matplotlib), Power BI (DAX, Power Query, semantic modeling, RLS), Tableau, Excel (Pivot Tables, XLOOKUP, VLOOKUP), SSRS

Data Engineering and Management: ETL/ELT pipelines, Snowflake, Data Warehousing, Dimensional Modeling, Source-to-Target Mapping, Data Validation, Data Integration, Report Automation

Project Management: Jira, Agile, Scrum, Stakeholder Collaboration, UAT Coordination, Requirements Gathering

Statistics and Modeling: A/B Testing, Hypothesis Testing, Regression Analysis, Time Series Analysis, Predictive Modeling, Classification and Clustering, Explainable AI (SHAP)

PROFESSIONAL EXPERIENCE

Research Data Analyst, University of South Florida - Remote | Jan 2026 - Present

- Analyzed 150K+ CHF inpatient admissions across 4K+ U.S. hospitals using Python and developed a multidimensional hospital financial risk index (COFRI), enabling data-driven evaluation of clinical, operational, and financial variables.
- Developed machine learning classification models on HCUP NIS datasets to execute Predictive Modeling and Advanced Analytics, achieving a 0.769 AUC to flag high-risk facilities.
- Identified emergency admission rate, patient severity, and diagnosis count as the top predictors of financial risk using SHAP explainable AI, reframing financial risk as a downstream signal of upstream clinical and operational patterns.

Graduate Teaching Assistant, University of South Florida - Tampa, FL | May 2025 - Dec 2025

- Mentored and guided 130+ graduate students in SQL, Python, Tableau, and Power BI across three business analytics courses, resulting in a 4.5/5.0 satisfaction rating and improved project performance.
- Guided students in building Power BI and Tableau dashboards, implementing SQL-based reporting logic, and performing exploratory data analysis for BI use cases.

Senior Data Analyst, Accenture - Bengaluru, India | Jan 2022 - May 2023

Client: Global fashion retailer

- Built Power BI dashboards and the underlying semantic model (Power Query, DAX) using POS and inventory data, delivering insightful KPIs to planning and operations teams.
- Developed a scalable reporting data layer and standardized KPI definitions across dashboards, establishing a single source of truth for operations, planning and finance teams.
- Analyzed historical sales and inventory data using SQL and Power BI to identify seasonal demand trends, improving replenishment planning decisions and reducing stockouts by 15%.
- Performed data reconciliation and testing of key metrics and KPIs using SQL and Excel that reduced post-release reporting defects by ~30%.

Associate Analyst, Software AG - Bengaluru, India | Mar 2020 - Dec 2021

Client: Tier-1 global bank

- Worked on building end-to-end data lineage and documenting source-to-target mapping to support compliance reporting for a Tier-1 bank.
- Authored functional specifications and business rule documentations using Excel and Visio; coordinated cross-functional UAT and go-live with compliance stakeholders, delivering a centralized, auditable reporting trail.
- Designed data quality checks that traced, validated, and flagged data integrity issues, thereby keeping compliance reporting accurate.
- Managed Jira sprints and task allocation to build analytics modules, ensuring on-time delivery of business performance metrics.

Analyst - Data and Integration, Accenture - Bengaluru, India | Mar 2016 - Feb 2020

Client: Global pharmaceutical, UK utilities provider

- Translated business requirements into technical data specifications and source-to-target mappings, driving integration workflows across SAP HANA, IBM Maximo, Ariba, and CRM platforms.
- Developed an automated error-alert and monitoring services that proactively flagged data and integration failures, reducing manual log monitoring by ~30 minutes/day per team member and improving incident response.
- Built reusable validation and reporting templates using Excel, standardizing data-quality checks and testing consistency across analyst teams.
- Mentored 7+ junior analysts in SQL, data validation, requirements documentation, and UAT; led knowledge-transfer sessions on integration patterns and client domain context.

EDUCATION

Master of Science, Business Analytics | Aug 2024 - Dec 2025

University of South Florida - Florida, USA

Bachelor of Technology, Computer Science and Engineering | Jul 2015

Sikkim Manipal University - Sikkim, India

PROJECT EXPERIENCE

U.S. Crime & Socioeconomic Trends Analysis (Python, Tableau) | Dec 2025

- Analyzed FBI UCR and U.S. Census data (2010-2019) across 50 states using Python for data cleaning, normalization, and feature engineering to enable consistent cross-state and temporal analysis.
- Built interactive Tableau dashboards visualizing socioeconomic factors against crime trends; identified that property crime correlates strongly with low-income states while unemployment shows weak correlation with violent crime.

Movie Investment Analytics & ROI Prediction (Oracle SQL, Tableau) | Jun 2025

- Designed a multi-dimensional data warehouse and built OLAP schemas for 45K+ movies and 20M+ user ratings to perform SQL analysis based on users, time-period, region, genre ROI, audience sentiment, and release trends.
- Built an interactive Tableau dashboard visualizing genre performance, ROI, and budget-to-revenue relationships to support data-driven film investment and content strategy decisions.